

Manea Abdullah Al-Awbathani

SYSTEMS ARCHITECT · FULL-STACK ENGINEERING LEAD · AI & SOFTWARE STRATEGIST

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Systems architect and full-stack lead who ships production platforms end to end — multi-tenant SaaS, real-time telehealth, and AI-native industrial systems across healthcare, compliance, and energy.

Currently building Aip's software and AI function from the ground up: cloud architecture standards, MLOps governance, and LLM, agentic, and RAG integration into ERP, MES, and SCADA stacks.

Co-founder and CTO who writes the architecture and builds the team to scale it — three-plus years turning hard enterprise requirements into systems that hold up in production.

SKILLS

AI & Machine Learning — LLMs (OpenAI GPT, Claude, LLaMA) · Agentic AI · RAG pipelines · Prompt Engineering · Vector DBs (Pinecone, Weaviate) · MLOps · EDA

Backend, APIs & Real-Time — NestJS · Express.js · Node.js · FastAPI · REST · GraphQL · Swagger/OpenAPI · Socket.io · WebSockets · event-driven architecture · BLE data streaming

Frontend — Next.js (15/16) · React (18/19) · Vue.js · Quasar · Tailwind CSS · Radix UI · Framer Motion · TipTap · Zustand · TanStack Query · React Hook Form · Zod · Chart.js · Recharts · D3.js

Data & Storage — PostgreSQL · MySQL · MongoDB · Redis · TypeORM · Prisma

Cloud & DevOps — AWS (EC2, S3, RDS, Lambda) · Docker · CI/CD · multi-tenant SaaS · Cloudinary

Languages — TypeScript · Python · JavaScript (ES2022+) · PHP · PL/SQL · Bash

Security & Auth — JWT · Passport.js · bcrypt · Helmet · RBAC · rate-limiting · OWASP practices

Testing, Integrations & Docs — Jest · Supertest · Testing Library · SendGrid · Twilio · Nodemailer · Winston · SRS/SDD/STR/STD

EXPERIENCE

Head of Software & AI — Aip

Jan 2026 – Present · Riyadh, KSA · Full-time, On-site

Owns the company's entire software and AI charter across enterprise and industrial systems — architecture, governance, and team.

- Sets cloud architecture standards and MLOps governance spanning every AI and software system, from data pipeline to production deployment.
- Architects AI-powered solutions that embed LLMs (OpenAI, Claude, LLaMA), agentic AI, and RAG pipelines directly into ERP, MES, CMMS, and SCADA products.
- Builds the internal engineering team from the ground up and converts executive priorities into production-ready systems.

Co-Founder & CTO — ISOPluss

Apr 2025 – Present · Riyadh, KSA · Concurrent, Part-time

Co-founded and architected an enterprise B2B SaaS digitizing ISO 22000 and HACCP compliance for global clients.

- Designed a multi-tenant SaaS architecture (NestJS, Next.js, PostgreSQL, Redis, TypeORM) with JWT/RBAC security and a fully documented OpenAPI surface.
- Built a schema-driven engine that generates digital workflows, document control, audit management, corrective actions, and analytics dashboards from configuration.
- Owns cloud infrastructure, CI/CD, DevOps automation, and production-grade data security end to end.

Software Engineer & Technical Lead — Me'Kaaz

Jul 2025 – Jan 2026 · Riyadh, KSA · Contract, On-site

Architected and delivered MENA's first BLE-smartwatch telehealth platform.

- Engineered real-time cloud infrastructure hitting <3s vitals latency at 99.9% uptime across B2C patient apps and B2B hospital systems.
- Cut branded hospital deployment to under one hour — dashboards, ward heatmaps, and automated clinical alerts included — with patient onboarding in under 60 seconds.
- Directed a cross-functional design, engineering, and QA team through regional rollout under NCA/MOH compliance, owning knowledge transfer and mentoring.

Assistant Development Manager — MYCES SDN BHD

Jul 2024 – Dec 2024 · Malaysia · Full-time, Hybrid

Delivered enterprise systems across the agriculture, dam-safety, and energy sectors.

- Shipped 3 major platforms (AgroFarm, Dam Monitoring, EMS) from requirements to MVP, compressing delivery timelines 20% with Agile and a reusable component library.
- Authored full SRS, SDD, and STR/STD documentation that cleared regulatory compliance reviews with zero audit flags.
- Mentored junior developers and established post-launch continuity for delivered systems.

Software Engineer — MYCES SDN BHD

Dec 2023 — Aug 2024 · Malaysia · Full-time, On-site

Built upgrades and features for EMARS, an energy-monitoring SaaS serving 100+ enterprise facilities.

- Reduced energy waste 17% across 100+ client facilities by shipping threshold-based alerting and tree-map analytics dashboards.
- Lifted user engagement 22% with responsive mobile and web interfaces, and hardened data security through RBAC.
- Stack: PHP, MySQL, Vue.js, Quasar, NestJS, TypeORM, D3.js. Started as a Full-Stack Intern (Oct–Nov 2023).

SELECTED PROJECTS

GavelMarket — Real-Time Online Auction Platform

Part-time · Oct 2025 — Present · Saudi Arabia (Remote)

High-concurrency online auctions demand fair, real-time bidding without dropping a single transaction.

Architected and built the full stack — Express.js 5, Prisma, PostgreSQL, Socket.io, JWT, Cloudinary, SendGrid, Twilio, node-cron, Winston, Zod (TypeScript) on the backend; Next.js 16, React 19, Zustand, TanStack Query, Framer Motion, Radix UI, React Hook Form, Tailwind on the frontend.

Tuned for high-concurrency bidding with timer-based lifecycle events, rate-limiting, structured logging, and data-integrity guarantees across secure payment flows.

COVITA — Multi-Vendor Coffee Marketplace (B2B/B2C)

Contract · Feb 2026 — Present · Riyadh, KSA (Remote)

The Saudi and Gulf coffee market needed one platform spanning wholesale, retail, and subscription commerce.

Led end-to-end architecture and development — vendor onboarding, product catalogs, order management, payment integration, and mobile/web apps.

Scaled to a multi-stakeholder ecosystem combining B2B wholesale, B2C retail, and subscription commerce on cloud infrastructure.

HealthWatch — Telehealth Platform (Me'Kaaz, Riyadh)

Hospitals lacked real-time remote vitals monitoring tied directly into clinical workflows.

Engineered the cloud architecture for a real-time ecosystem connecting BLE smartwatches, patient mobile apps, and a multi-tenant hospital portal.

Delivered <3s vitals latency with automated clinical alerts, ward heatmaps, and role-based doctor–patient interactions; NCA/MOH compliant and scaling into B2B hospital onboarding.

ISOPLUSS FSMS — Food Safety Management System (ISOPluss, Riyadh)

ISO 22000 and HACCP compliance is manual, paper-heavy, and hard to audit at scale.

Architected a modular enterprise SaaS (NestJS, Next.js, PostgreSQL, Redis) with a schema-driven workflow engine and a fully Swagger-documented API.

Runs as multi-tenant cloud infrastructure covering dynamic document control, audit tools, objectives tracking, corrective actions, and analytics.

AgroFarm — AgriTech Monitoring Platform (MyCES, Malaysia)

Farm operations were fragmented across manual, disconnected processes.

Built a farm-monitoring platform with real-time data, resource planning, automated pricing tools, and reporting.

Awarded the Gold Medal at the Johor Agriculture Department Innovation Competition 2024 — recognized as a national benchmark for agricultural digital innovation.

PUBLICATIONS

IEEE — Charity and Donation Tracking System Using Queue Structure (published research paper).

EDUCATION

Bachelor of Computer Science (Software Engineering) — University Malaysia Pahang, Pahang, Malaysia

Oct 2020 — Mar 2024

ACCREDITATION, CERTIFICATIONS & AWARDS

- **Saudi Council of Engineers** — Professional Accreditation Certificate, Specialist (Computer Science)
- **IBM** — Exploratory Data Analysis for Machine Learning
- **Gold Medal** — Johor Agriculture Department Innovation Competition 2024 (AgroFarm, MyCES)

Additional coursework: Quantum Computing (Saint Petersburg State University) · Blockchain Basics (University at Buffalo)